K782 — Q1 is COMPUTED, and it's honest: the natural geometry gives a smooth Gindikin Γ-ratio, NOT 1/2^g. 127/128 is contingent on the substrate being discrete. Casey's response — continuous geometry that TILES into a discrete code during cooling — inverts the mainstream and is the right, decidable frame.

Keeper | 2026-07-20 | Lyra computed the decider (the actual radial Bergman integral, not a lattice model). It came back NOT confirming 127/128 — the most valuable thing that happened today. + Casey's continuous-tiles-into-discrete hypothesis. Web-grounded (spacetime-from-QECC). This is the second de-inflation in two rounds; honored.

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1. The computed result (honor it first — it's the point)

Lyra did the real radial Bergman integral. The deficit comes out a smooth Gindikin Γ-ratio, not the discrete 1/2^g. Rank-1 disk anchor: $\sqrt{((n+1)/(n+\alpha+2))} \sim 1 - c/n$, a *continuous* function of the radial level; rank-2: a two-parameter Gindikin Γ, still smooth. Even at the naive "level 127," the smooth deficit is ~0.015, not 1/128 = 0.0078. So: - The geometry does NOT produce 127/128. It produces a Γ-ratio; 127/128 is just one point on a smooth curve. - The "128" is NOT derived by the continuous radial computation — it is the Reed–Solomon code truncation (discrete). Answer to round 6's sharp question ("128 derived or imposed?"): imposed by the code, not output by the geometry. - 127/128 is exact ONLY IF the substrate is discrete-fundamental. The continuous geometry alone gives a smooth family (the top's level is a free parameter, no prediction); the *code* pins the top to the maximal codeword (127/128, a prediction). So 127/128's content IS the discreteness.

This is a genuine de-inflation: 127/128 is not "one integral away from a theorem" — the integral was done, and it gives something else. 127/128 now rests on an unproven premise (fundamental/emergent discreteness), and the natural continuous computation disagrees with it.

2. The reframe: Q1 is about the substrate's NATURE (and it's the it-from-qubit question)

Not "one radial integral," but: is the substrate fundamentally the discrete code (geometry emergent), or fundamentally the continuous geometry (code emergent)? The two predict *different* deficits (1/2^g vs Γ-ratio) — so it's decidable in principle. The mainstream (Almheiri–Dong–Harlow, HaPPY): discrete code fundamental → continuous spacetime emergent ("spacetime emerges from a QECC").

3. Casey's hypothesis — and it INVERTS the mainstream

The substrate is continuous D_IV^5 geometry; during COOLING it undergoes a phase transition (crystallization/tiling) that forms discrete cells — the RS code EMERGES from the continuous geometry. The tile boundaries are "more solid" (domain walls / higher-energy barriers) keeping the substrate inside "fresh"; light and neutrinos are emitted at the boundaries. So: continuous before tiling, discrete after — the *reverse* of the mainstream (which has discrete-fundamental → continuous-emergent). This is a distinctive, bold BST claim, and it is exactly the right frame for Lyra's result: - It's consistent with the computed geometry: the natural computation IS continuous (a Γ-ratio); the discrete code is therefore *emergent* (a coarse-graining / error-correction on the continuous geometry), not fundamental. Casey's direction matches what Lyra found. - It's supported by Elie's $3\pi/128$: the Faraut–Koranyi Bergman norm $\|f\_(½,½)\|^2 = \Gamma(5/2)^2/\Gamma(5) = 3\pi/128 = N\_c\cdot\pi/2^g$ — the 128 falls out of the continuous Bergman integral, not a modeling assumption. So the code's cell-count 2^g has a *geometric origin*: the continuous geometry already "carries" a 128, which the tiling realizes as cells. (But $3\pi/128$ is a *norm*, not the *overlap* the deficit needs — it points to "128 has a geometric home," not to "127/128 is the gap.") - The mechanism = a phase transition (the search: CFTs/criticality arise at phase changes like superconductivity) — the tiling is the substrate's crystallization as it cools, exactly Casey's picture.

4. The decidable test (how the tiling either rescues or kills 127/128)

Casey's hypothesis is testable, not a rescue-by-fiat: - Does the tiling pin the top to the MAXIMAL codeword (127/128)? Naive "snap the continuous value (~0.985) to the nearest codeword" would give 126/128 (0.984), not 127/128 — so the tiling must specifically produce the *extremal/maximal* codeword, not the nearest. Show that it does (via the crystallization dynamics), or 127/128 doesn't follow. - Does the observed top mass discriminate? The pure-continuous Γ-ratio (deficit ~0.015 at level 127) → m_t ~171.5 GeV; the code (127/128) → 172.74; observed ~172.7. So the *data favors the code value* — meaning IF the top sits at the code's maximal codeword, it matches; the continuous geometry only matches by choosing the level (free parameter). The code's content = pinning the top to the extremal cell.

5. How to continue the investigation

1. ★ Model the tiling / crystallization phase transition of continuous D_IV^5 during cooling: does it produce a 2^g = 128-cell RS code, and does it pin the top to the *maximal* codeword (127/128)? This is the decidable core now.
2. ★ The geometric origin of 128 (Elie's $3\pi/128$): does the continuous Bergman geometry force a 2^g cell-count under tiling? Connect the norm's 128 to the code field size.
3. The tiling physics: boundaries as domain walls (solid, higher-energy); light + neutrinos emitted at boundaries; the "fresh interior." Model the emission.
4. The mainstream-inversion claim (continuous → discrete-emergent vs the standard discrete → continuous-emergent) is a distinctive, *publishable* BST position — and testable (Γ-ratio if no tiling, 1/2^g if tiled).

Honest tiers (twice de-inflated)

- Computed (real): the natural radial deficit is a Γ-ratio, NOT 1/2^g (Lyra). This is a genuine result — the decider ran and disagreed with the pretty number.
- 127/128: downgraded — a *conditional prediction contingent on the substrate being discrete/tiled*; the natural continuous computation gives a different (smooth) value. No longer "a lead one integral away."
- Casey's tiling hypothesis: a *testable* frame (does the tiling pin the top to the maximal codeword?), physically motivated, inverting the mainstream — the constructive path, not a rescue.
- Derived spine untouched: O = the vector, rank-1 → one mass, N_c quark, ceiling, m_e↔top, boundary-reach, angular = 1. Solid regardless.

— Keeper K782, 2026-07-20. Q1 COMPUTED: natural radial Bergman integral gives a smooth Gindikin Γ-ratio, NOT 1/2^g (Lyra) — 127/128 is not the continuous geometric answer; it's contingent on discreteness (imposed by the code, not output by the geometry). Reframe: Q1 = substrate's nature (discrete-fundamental vs continuous-fundamental) = the it-from-qubit question, decidable (Γ-ratio vs 1/2^g). Casey's hypothesis: continuous D_IV^5 TILES into a discrete code during cooling (boundaries=domain walls, light+ν emitted there) — INVERTS the mainstream (discrete→continuous). Consistent with Lyra's continuous result + Elie's $3\pi/128$ (128 in the Bergman norm = geometric origin of the cell-count). Decidable test: does the tiling pin the top to the MAXIMAL codeword (127/128) or the nearest (126/128)? 127/128 DOWNGRADED to a conditional-on-discreteness prediction. Derived spine untouched. See [[Keeper_K780_de_inflation_ratify_Lyra_catches_correct_my_own_framing_reanchor_on_Q1_2026-07-20]].

Sources

- Spacetime emerges from a QECC (continuum limit of a discrete lattice code); Almheiri–Dong–Harlow; HaPPY; CFT/criticality at phase transitions. Quanta: space-time as a QECC; Spacetime as a QECC? (ScienceDirect)